

Healthcare Metrics Project

Project Overview

- **Objective:** We need a unified view of hospital staffing and operational performance across all facilities to understand how nurse availability, workload, and patient volumes affect care quality and efficiency.

The system should provide clear insights into nurse-to-patient ratios, total working hours, and overtime trends by hospital and state. It should also help us track hospital occupancy, bed utilization, and identify where staffing levels are not aligned with patient load. We want to understand how staffing patterns impact outcomes such as readmissions and length of stay, and identify hospitals with high overtime or low staff-to-patient coverage.

All these insights should be available through an interactive dashboard to help management improve workforce planning, optimize costs, and maintain high-quality patient care across the network.

Step-by-Step Plan

Step 1: Source Data Explanation

1. **Task:**
 - Download the master CSV(**PBJ_Daily_Nurse_Staffing_Q2_2024.csv**) and 15 supporting files from Google Drive.
 - Verify the integrity of the downloaded files (e.g., check for missing or corrupt files).

All the files are present in the google drive link below
[Nursing Data Supporting CSV](#)

#	Column	Description
0	PROVNUM	Provider number, a unique identifier for each provider.

1	PROVNAME	Provider name, indicating the name of the healthcare facility.
2	CITY	The city where the provider is located.
3	STATE	The state where the provider operates.
4	COUNTY_NAME	Name of the county where the provider is located.
5	COUNTY_FIPS	Federal Information Processing Standard (FIPS) code for the county.
6	CY_Qtr	Calendar Year Quarter, indicating the quarter of the year.
7	WorkDate	The work date, typically in a numeric or timestamp format.
8	MDS Scensus	MDS (Minimum Data Set) census, likely indicating the number of patients.
9	Hrs_RNDON	Total hours worked by Registered Nurses (RN) on duty.
10	Hrs_RNDON_emp	Total hours worked by Registered Nurses employed directly by the provider.
11	Hrs_RNDON_ctr	Total hours worked by contracted Registered Nurses on duty.
12	Hrs_RNadmin	Total hours worked by Registered Nurse Administrators.
13	Hrs_RNadmin_emp	Hours worked by RN administrators employed directly by the provider.

14	Hrs_RNadmin_ctr	Hours worked by contracted RN administrators.
15	Hrs_RN	Total hours worked by all Registered Nurses.
16	Hrs_RN_emp	Hours worked by employed Registered Nurses.
17	Hrs_RN_ctr	Hours worked by contracted Registered Nurses.
18	Hrs_LPNadmin	Hours worked by Licensed Practical Nurse (LPN) administrators.
19	Hrs_LPNadmin_emp	Hours worked by employed LPN administrators.
20	Hrs_LPNadmin_ctr	Hours worked by contracted LPN administrators.
21	Hrs_LPN	Total hours worked by Licensed Practical Nurses.
22	Hrs_LPN_emp	Hours worked by employed Licensed Practical Nurses.
23	Hrs_LPN_ctr	Hours worked by contracted Licensed Practical Nurses.
24	Hrs_CNA	Total hours worked by Certified Nursing Assistants (CNAs).
25	Hrs_CNA_emp	Hours worked by employed Certified Nursing Assistants.
26	Hrs_CNA_ctr	Hours worked by contracted Certified Nursing Assistants.

27	Hrs_NAtrn	Total hours worked by Nursing Assistants in training.
28	Hrs_NAtrn_emp	Hours worked by employed Nursing Assistants in training.
29	Hrs_NAtrn_ctr	Hours worked by contracted Nursing Assistants in training.
30	Hrs_MedAide	Total hours worked by Medication Aides.
31	Hrs_MedAide_emp	Hours worked by employed Medication Aides.
32	Hrs_MedAide_ctr	Hours worked by contracted Medication Aides.

Step 2: Initial Data Analysis

1. Task:

- Analyze the master CSV and supporting files to understand the schema, relationships, and data quality issues.
- Identify key columns for joins and insights.

Information about the columns are given [HERE](#)

- Initial report on missing values, duplicates, and outliers
- Use Python (Pandas/NumPy) OR SQL for data exploration.
- Create basic visualizations (e.g., histograms for missing values).

Step 3: Define Data Engineering Pipeline/Architecture

1. Task:

- Design a pipeline architecture for ingesting, transforming, and storing data.

- Design an architecture / system design based on below requirements. **Complete below quick course related to data lake before moving ahead with your system design:**

📺 Data Lake Fundamentals

Consider your source as Google Drive i.e you need to ingest data from google drive incrementally.

Present the architecture for Subject Matter Expert (SME) approval. Stick to AWS SERVICES ONLY

Move head once clearance from the SME

Deliverable:

- Pipeline architecture diagram and explanation.
- Create a solution design document.
- Written approval from the SME.
- Use tools like draw.io, miro, etc for diagrams.

Step 4: Build the Data Pipeline

Deliverable:

- Working pipeline that can handle raw data ingestion to data warehouse.

Step 5: Define and Calculate Metrics

Disclaimer: You don't have data to solve all the below metrics. You need to understand the data and identify which among them are calculatable and calculate those.

We are looking at 3-5 metrics to be solved

1. Task:

- Calculate the key metrics like the ones shown below necessarily not the mentioned you need to come up with the metrics based on the available data:

Staffing Metrics

- Average nurse-to-patient ratio by hospital, state, and department.
- Total hours worked by nurses per hospital, state, and month.
- Percentage of nurses working overtime.
- Number of shifts per nurse (average and median) by hospital and state.

2. Facility Metrics

- Hospital occupancy rate trends over the past year (monthly/quarterly).
- Bed utilization rates by hospital and department.
- Comparison of staffing levels vs. bed occupancy rates.
- Top 10 hospitals with the highest patient throughput.
- Facilities with the lowest staffing levels compared to patient load.

3. Quality Metrics

- Patient satisfaction scores by hospital (if data is available).
- Average length of stay (ALOS) by department and state.
- Readmission rates within 30 days by hospital, state, and diagnosis category.
- Patient-to-nurse complaint ratio (if available).
- Correlation between nurse staffing levels and readmission rates.

4. Cost Metrics

- Total payroll costs for nurses by hospital and state.
- Average cost per patient stay by hospital and state.
- Cost of overtime hours as a percentage of total payroll costs.
- Comparison of hospital revenue vs. payroll expenses (if data available).

5. Operational Metrics

- Shift utilization rates by time of day (morning, afternoon, night).
- Peak staffing hours for each hospital and department.
- Ratio of permanent staff to temporary/contract staff.
- Trend analysis of nurse attrition rates (if applicable data exists).

HINT:

SQL queries and Python scripts for calculating metrics.

Step 6: Build the Dashboard

1. **Task:**

Design an interactive dashboard showcasing the calculated metrics and insights.

Deliverable:

- ☐ Completed dashboard with:
 - Staffing insights (e.g., nurse-to-patient ratio).
 - Facility performance metrics (e.g., occupancy rates).
 - Visualizations (e.g., bar charts, heatmaps, trends over time).

Please go through this link for working on Streamlit for Visualization:

[Streamlit Tutorial](#)

Step 7: Submission and Review

1. **Task:**

- Submit all deliverables, including:
 - Pipeline documentation.
 - Code files (ETL scripts, SQL queries).
 - Data dictionary.
 - Final dashboard.

2. **Instructions:**

- Ensure all files are properly named and documented.
- **Include a short video or presentation explaining the work.**
- Explain '**WHY**' behind each and every tech stack used in your design.

Questions Need to Answer

1. What is the relationship between nurse staffing levels and hospital occupancy rates?
2. Which hospitals have the highest overtime hours for nurses?
3. What are the average staffing levels by state and hospital type?
4. What trends can you identify in patient length of stay over time?

If you have chosen any specific metrics you can have your own conclusion notes.

Thank you.